

HTR and OCR for Armenian, Byzantine Greek, Coptic and Syriac

Consulting Project with the NOMOS Project

Astha Gupta Krrish Agarwalla
Project Partner: Prof. Dr. Zachary Chitwood
Supervisor: Dr. Andreas Bender



Department of Statistics, LMU Munich

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- 4 Model and Experimental Setup
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Introduction



Syriac

Estrangela

Serto

Madnhaya

right to left,
22 consonant letters;
vowels marked around them

Coptic

Sahidic

Bohairic

Fayyumic

Akhmimic

Lycopolitan

left to right,
24 Greek-derived +
6 to 7 Demotic-derived

Armenian

erkatagir

bolorgir

notrgir

shghagir

left to right,
36 original + 2
later letters

Byzantine Greek

majuscule

minuscule, from the 9th century

left to right,
24 letters; breathings,
accents and ligatures

Four traditions, and each one is several scripts.

These are the manuscripts



Greek, Grec 1360

dated 1351, law and canon law

5 pages, 204 lines



Syriac, Chapter 4

watermark over the ink

18 pages, 504 lines



Coptic, Borg. copt. 109

White Monastery fragments

33 pages, 2,823 lines



Armenian, MS_UCLA_MS

ff. 64v to 73v, two columns

34 pages, 1,738 lines

BnF; BAV Borg. sir. 82 and Borg. copt. 109; MS_UCLA_MS page 1

Old, faded and torn pages



Brown water stains
swallow text



Blurred microfilm, dark margin
blotches



Stained corners,
ghost text behind



Parchment torn away
mid-column

A person has to read these before anything else can.

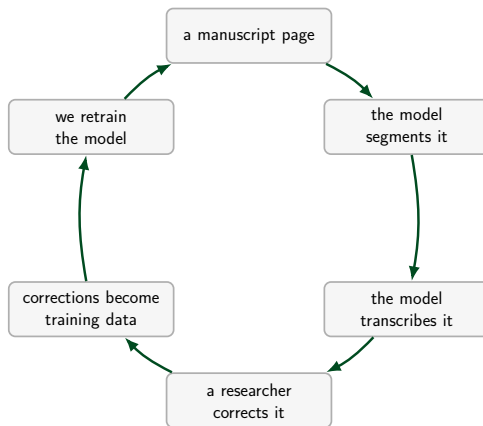
	Public handwriting models	Best off the shelf
Byzantine Greek	kraken and Transkribus models	32.7 %
Syriac	Beth Mardutho models	14.7 % the exception
Armenian	almost nothing handwritten	printed models only
Coptic	printed Coptic only, no handwriting	80.8 %

No public model reads Armenian, Greek or Coptic handwriting well. Syriac is the exception.

CER is the share of characters wrong.

Public models^{1-3,5}; systems we ran^{4,7,11,12,18,19} plus party and Qwen3-VL.

A corpus, a transcription model that runs on our server, and a platform where the researchers correct.



Corrections are designed to train the next model.

5,400+

lines of curated data with expert text, in four scripts

under 20 M

parameters per model: resource efficient, and they run very well on a CPU

public

platform and transcription models, corrections flow back

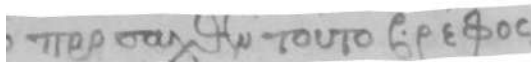
02

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Data: Challenges and Curation

- The challenges, script by script
- The corpus





Second word from the right, *touto*: the label had Latin
o typed twice

O

o

K

κ

Latin o

Greek omicron

Latin K

Greek kappa

U+006F

U+03BF

U+004B

U+039A

Same shape, different code point: 152 rows

λόγος.

full stop

U+002E

λόγος·

Greek ano teleia

U+0387

λόγος;

Greek question mark

U+037E

Two dot-like marks plus the Greek question mark. We unified each mark.

Coptic: lines too close together

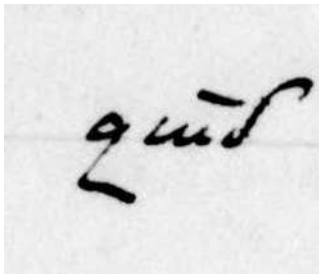


Two columns, tight lines



Boxes merge and split

Tight lines cut badly, and a bad cut cannot be read.



Zoom: the pativ abbreviation mark, U+055F

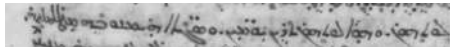
On the page	As it was typed
գամ	գամ[ենայն]
pativ overline: U+055F	bracketed expansion

The label preserves the page's pativ mark (U+055F); the bracketed expansion is shown only for comparison.

The label must say what the page says.

Calfa/BnF Dulaurier

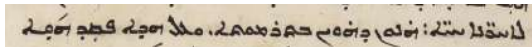
Syriac: two densities, many dots



West Syriac (Serto): sparse, narrow



East Syriac (Madnhaya): dense, pointed



Dots that change the reading

ܕ ܕ

dot below: *d*; dot above: *r*

ܒ ܒ

dot above: hard *b*; below: *v*

ܡܠܟܐ ܡܠܟܐ

two dots above: king → kings

ܡܠܟܐ ܡܠܟܐ

vowels: East dots, West Greek signs

One script, two traditions: the density difference is large, and dots decide words.

West line: OeNB Cod. Syr. 1 (Vienna, Serto); east line: Alqosh 65 (East Syriac); dots: Vat. Borg. sir. 82

script	corpus	train	validation	test	total lines
Armenian	pretraining corpus	1,807	297	282	2,386
	finetuning corpus	1,141	119	120	1,380
Greek	pretraining corpus	18,239	2,298	2,274	22,811
	finetuning corpus	164	19	21	204
Syriac	pretraining corpus	14,499	1,791	1,799	18,089
	finetuning corpus	2,463	331	335	3,129
Coptic	pretraining corpus	none			
	finetuning corpus	2,258	282	283	2,823

Greek finetuning is 204 lines. Coptic has no pretraining corpus.

Pretraining data: whatever we can find

every corpus



Greek inscription and painted text



Narrow-column manuscript page



Illuminated, pointed Syriac page

Pretraining widens the script coverage; finetuning teaches the target manuscript.

Source pages collected for pretraining; illustrative examples

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Segmentation

- From a page to a training line
- Cutting the page into lines: kraken
- Measuring segmentation
- Pairing lines with text, and checking them





Bibl. lat. 1360 / Bibliothèque nationale de France, Département des Manuscrits, Paris, 1360

grec1360_p7_lines.txt

- 1 ποιεῖν αὐτὴν ἀπαιτούμεθα, ὥς ὑποποιὸν καὶ θεοφιλῇ καὶ σωτήριον.
- 2 Εἴθ' οὕτω καὶ τάδε διανοείσθω· καὶ τοῖς λογισμοῖς περιστρε
- 3 φέτω· πρὸς τοῦ Κυρίου καὶ Σωτῆρος ἡμῶν εἰρημένα, τοῦ ἀληθινοῦ
- 4 Θεοῦ καὶ δικαίου τῶν ὅλων κριτοῦ· Μὴ κρίνετε κατ' ὄψιν· ἀλ
- 5 λὰ τὴν δικαίαν κρίσιν κρίνετε· καὶ μακάριοι οἱ πεινῶντες
- 6 καὶ διψῶντες τὴν δικαιοσύνην, ὅτι αὐτοὶ χορτασθήσονται.

... 51 lines for this page

Per-line text from the researchers

A manuscript page image

BnF Grec 1360, opening 7



A scan, many lines



One cut line, one line of text

Page level OCR and HTR need a lot of data.

We do not have it.

scan → lines → pairs → training

BnF Grec 1360



One Coptic page, two segmenters: kraken BLLA left, Orli right

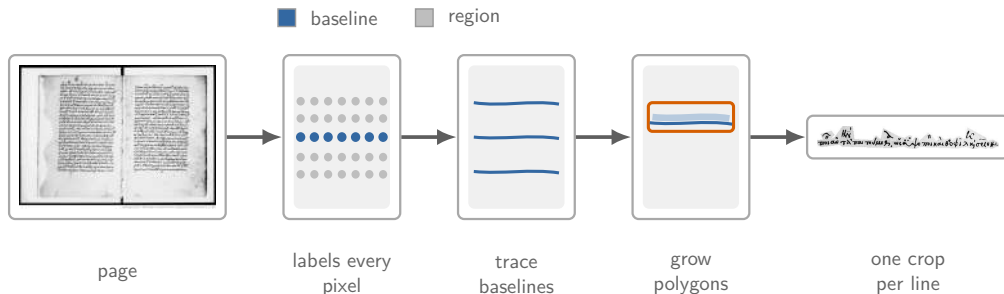
Running today. Checked by eye, looked good.

No gold segmentation set existed yet, so nothing was measured.

Fast, CPU only, in the platform, with baseline and polygon¹¹.

Small: the kraken BLLA segmenter has 1.26 M parameters, a 5 MB file.

How kraken segments a page



kraken BLLA (baseline layout analysis) returns a baseline and a polygon¹¹.

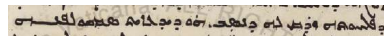
BnF Grec 1360, opening 7

polygon: the outline around one line of writing



baseline: the line the letters sit on

A detail of one Syriac line, annotated



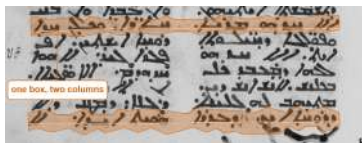
The crop: all the transcription model
ever sees

kraken BLLA (baseline layout analysis) returns both¹¹.

BAV Borg. sir. 82



Sloping lines split in two



One box over two columns



Faded ink: a column is lost

When segmentation fails, pairing fails.

Greek microfilm opening; HMML SMMJ 00036; Naples IB.10.10-27



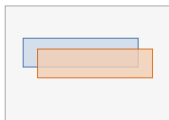
Page edges and corners



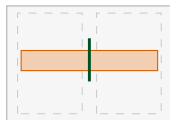
Library banners and footers



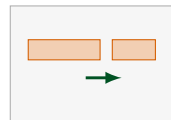
Slivers and tiny boxes



Overlaps: trim or drop the copy



One box over two columns: split



Fragments on one baseline: merge

Then renumber the boxes column by column, so reading order is right.

The tool suggests, a person confirms.

Where kraken failed, we fixed the boxes by hand



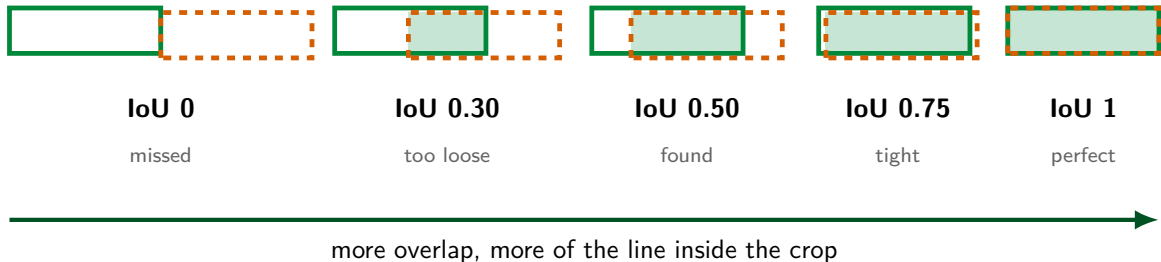
kraken found 41 of 63 lines



Krrish drew or reshaped 37 by hand

Hand work came from kraken's misses. The correction loop on the platform saved us hours of it.

Higher IoU means a better box



$\text{IoU} = \text{shared area} / \text{area covered by both boxes}$: 0 is no overlap, 1 is the same box.

The transcription model only sees the crop. **Higher IoU, more of the written line reaches it.**

too loose, IoU = 0.30



found and roughly right, IoU = 0.50

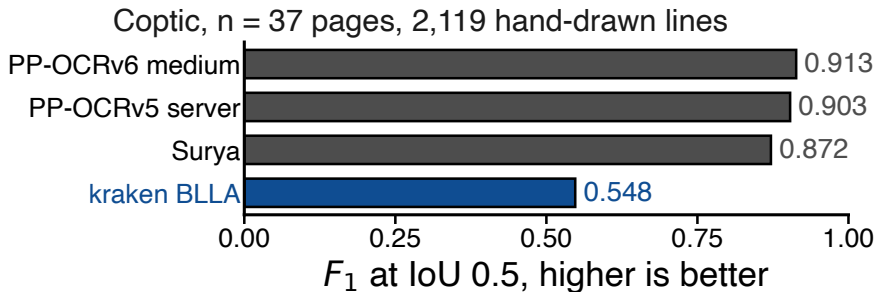


tight, IoU = 0.75



— gold polygon - - - predicted box ■ shared area

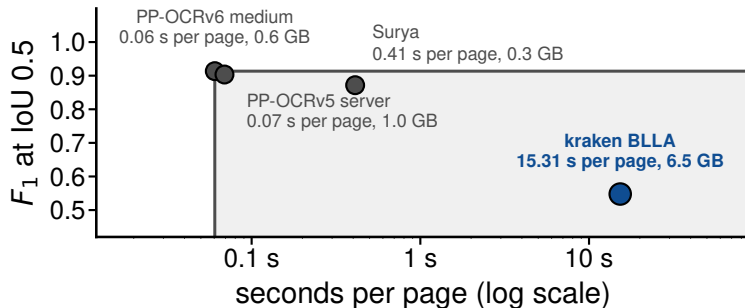
At **0.5** the line is found and roughly right, and still reads. A missed line is lost text.



PP-OCRv6 medium F_1 **0.91**, kraken BLLA 0.55

Planned: replace kraken BLLA with PP-OCRv6 medium.

● runs on the platform today ● off the shelf, measured here — nothing beats this line on both axes



37 Coptic pages, 2,119 hand-drawn gold lines; GB is peak GPU memory, batch 1

Nothing we measured is both faster and better than PP-OCRv6 medium.

The same page, three segmenters



kraken BLLA

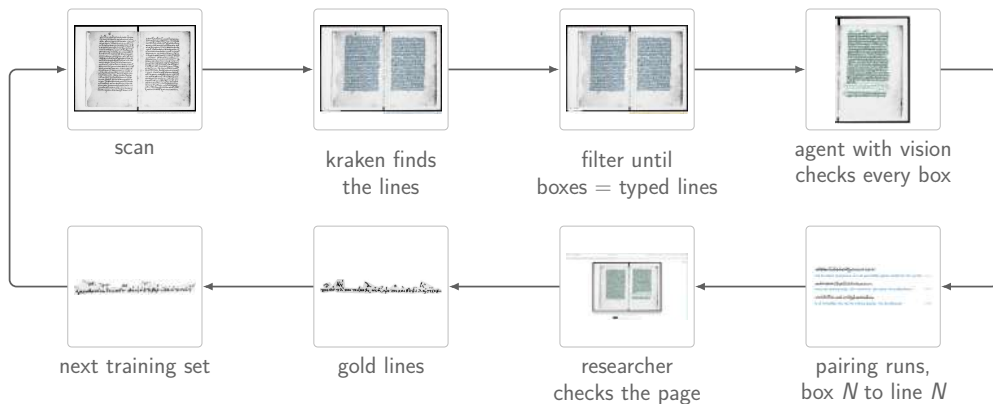


PP-OCRv6 medium



Surya

Green matched a gold line, purple is half a line, red is a box with no writing in it.



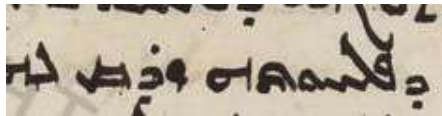
Nothing becomes gold data without the last step.



263 Byzantine Greek



1,738 Armenian



647 Syriac



2,823 Coptic

5,471 lines paired. About 90 to 95% automatic; the rest still needs a person.

BnF Grec 1360; MS_UCLA_MS 72; BAV Borg. sir. 82; BAV Borg. copt. 109

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Model and Experimental Setup



Metric 1: character error rate (CER)



reference ƿ ƿ ƿ o ƿ þ ƿ ƿ . ƿ j n ƿ ƿ

prediction ƿ ƿ q o ƿ þ ƿ ƿ ƿ j n ƿ ƿ

CER = edits / reference characters = 4 / 15 = **0.27**

edits: 3 substitutions + 1 deletion

MS_P_172





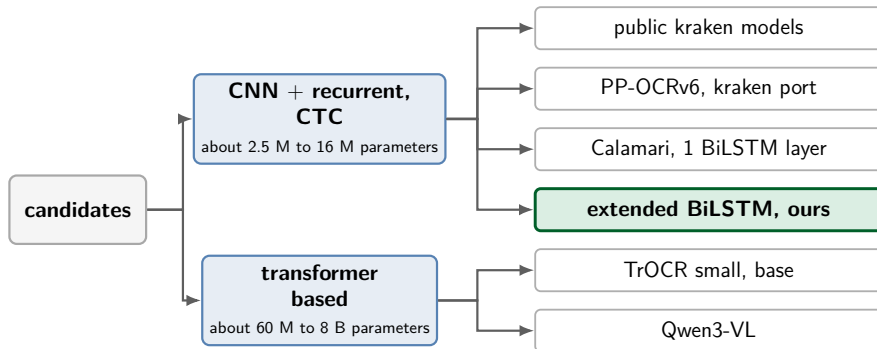
reference ƿ ƿ ƿ ƿ ƿ ƿ ƿ . ƿ ƿ ƿ ƿ ƿ

prediction ƿ ƿ ƿ ƿ ƿ ƿ ƿ ƿ ƿ ƿ ƿ ƿ

exact match = lines with zero errors / lines = 0 / 1 = **0.00**

The line is either perfect or it is not; useful when a researcher wants to know how many lines need no correction at all.

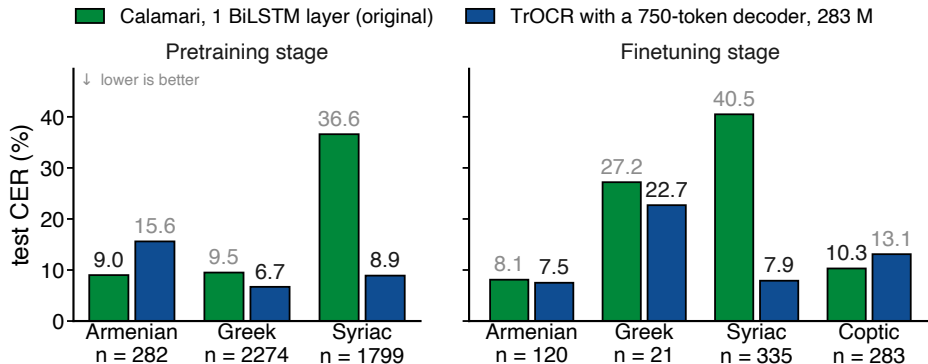
MS_P_172



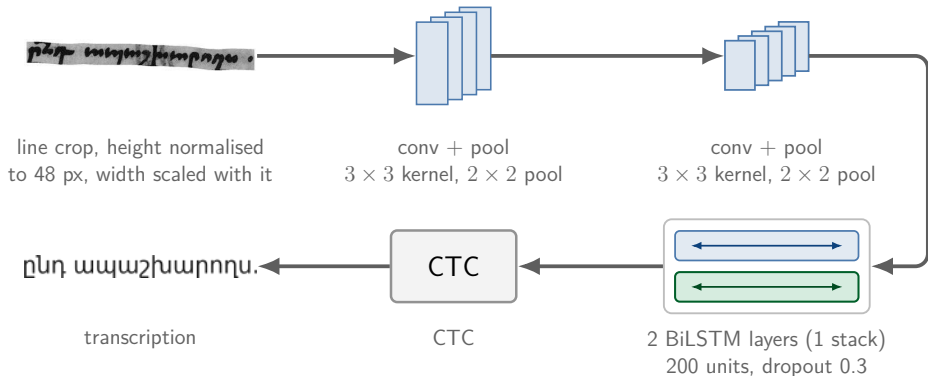
kraken¹¹ Calamari¹⁹ PP-OCR⁷ PP-OCRv6⁴ TrOCR¹²

Two criteria: error on the finetuning test split, and it runs on a laptop CPU.

The extended BiLSTM is the transcription model in production today.



The original Calamari is already close or ahead on Armenian, Greek and Coptic, and about **180 times smaller**. Syriac is the gap.



Height 48 to 12 before the BiLSTM. Small, runs on a laptop CPU.

40/60 filters for Greek, Syriac and Coptic, 60/80 for Armenian.

Calamari¹⁹ CRNN¹⁷ CTC⁸ LSTM⁹

Pretraining

Data	script corpus, mixed sources
Train lines	Arm. 1,807 Gr. 18,239 Syr. 14,499 no Coptic corpus
Starts from	random initialisation
Epochs	150
Learning rate	10^{-3}
Batch size	32
Augmentation	5 copies per line
Checkpoint	best validation CER

Finetuning

[▶ the full recipe](#)

Data	the target manuscript
Train lines	Arm. 1,141 Gr. 164 Syr. 2,463 Copt. 2,258*
Starts from	the pretrained checkpoint
Epochs	100
Learning rate	10^{-4}
Batch size	32
Augmentation	5 copies per line
Checkpoint	best validation CER

Scored once on the finetuning test split: Armenian 120, Greek 21, Syriac 335, Coptic 283 lines.

Both stages: seed 1111, one run, no intervals; page-disjoint splits; weight decay 0.01, warmup 9 %, EMA 0.99; line height 48, dropout 0.3. Full protocol in the appendix.

*Coptic has no pretraining corpus and is trained single-stage with the finetuning recipe.

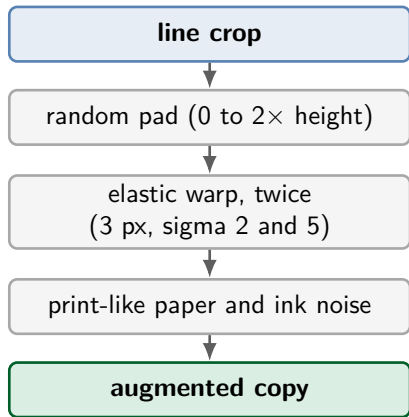
05

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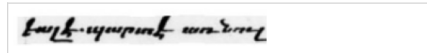
Results

- Making Calamari better
- The final models

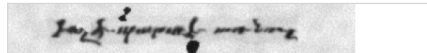




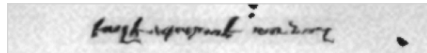
original



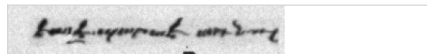
copy 1



copy 2



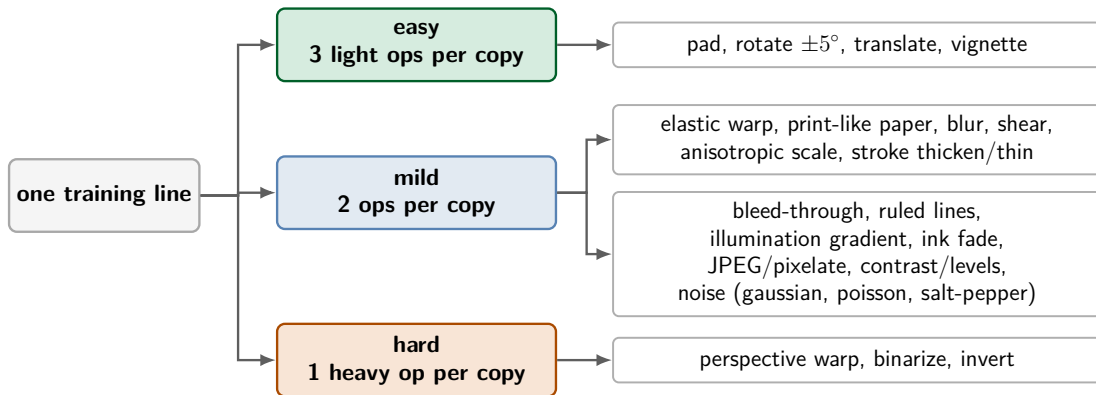
copy 3



Original recipe, three copies of the same line

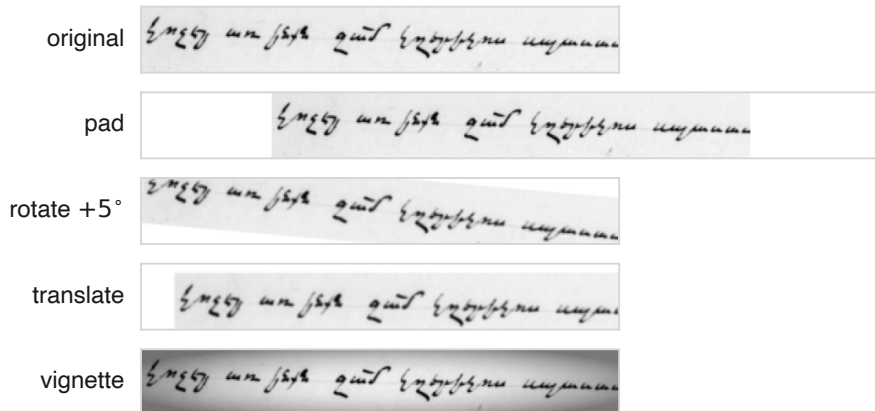
One fixed pipeline for every copy: the same kind of change, every time.

Our augmentation: three tiers, 22 operations



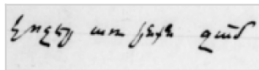
Light changes keep the line readable; heavy ones change the domain. Syriac also gets a width jitter.

Easy tier: the line stays the same

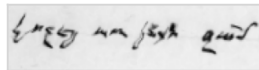


Every panel is one operation on its own, at the strong end of its own range.

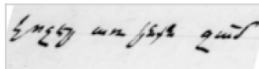
original



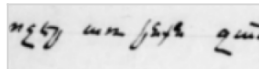
elastic warp



shear



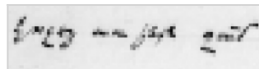
anisotropic
scale



blur



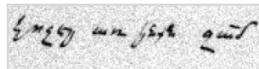
pixelate



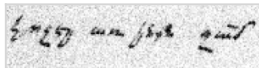
levels
(equalize)



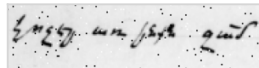
gaussian
noise



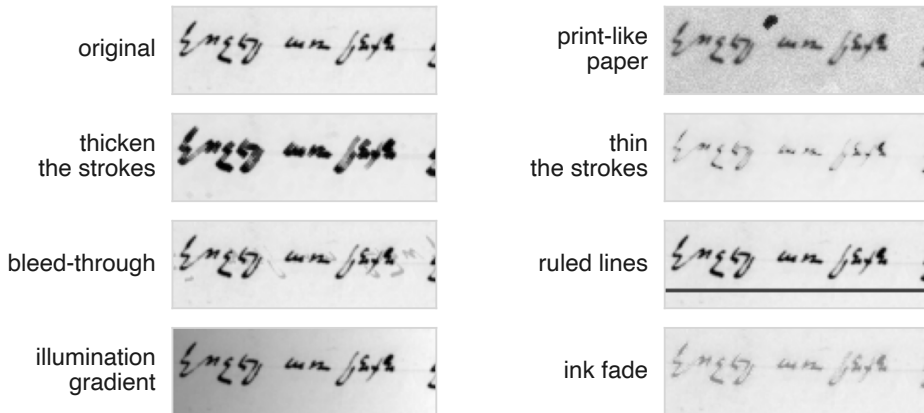
poisson
noise



salt and
pepper

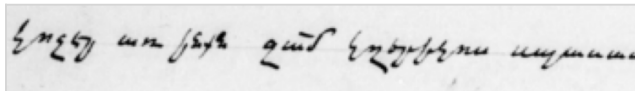


The strokes bend, blur and pick up grain, and the writing survives all of it.

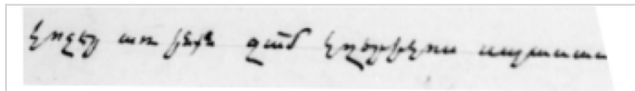


The page changes around the writing, and the strokes stay where they are.

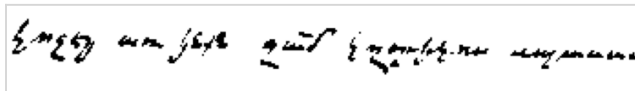
original



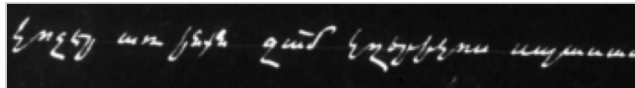
perspective
warp



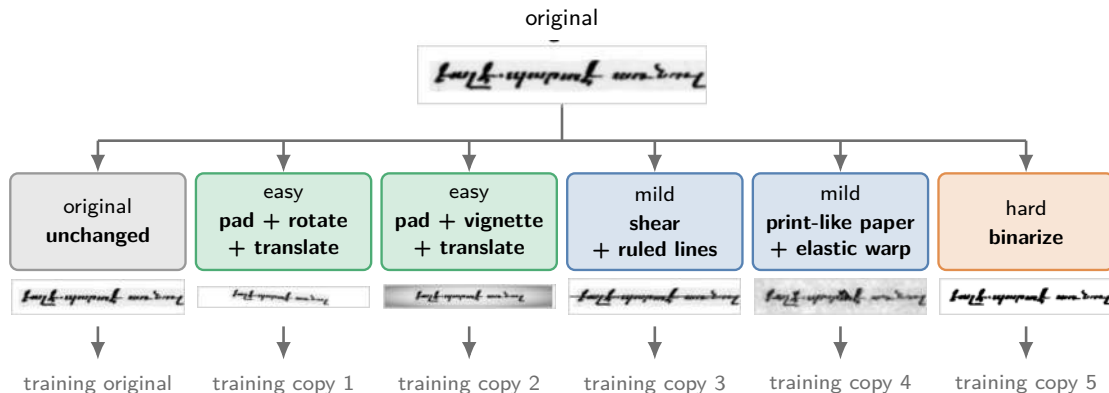
binarize
(Otsu)



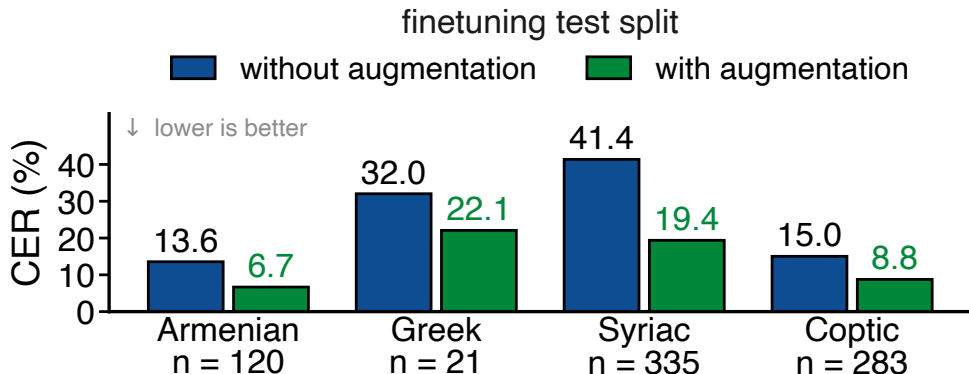
invert



What changes here is the kind of image, not the handwriting in it.



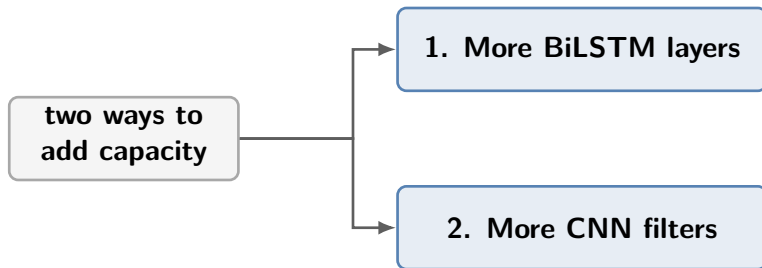
Every epoch sees the original plus five fresh copies;
the tier is fixed per slot, the operations are drawn anew.

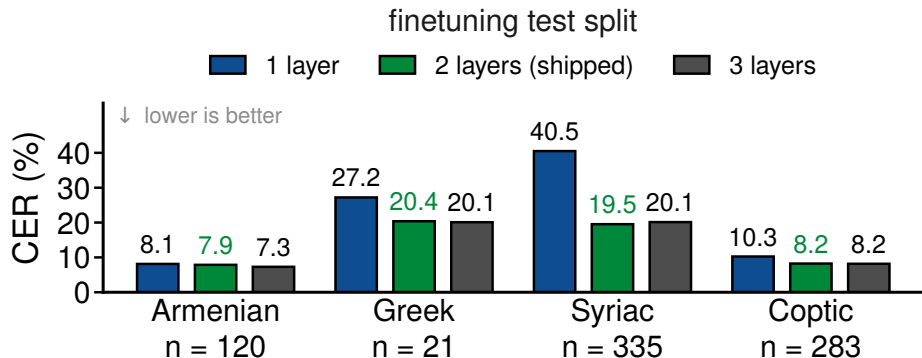


21 lines: indicative

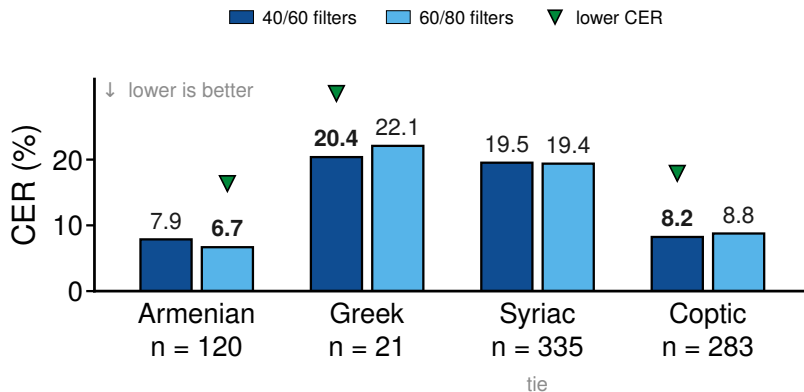
Augmentation lowers the error on all four scripts.

Augmentation has done its part. The original architecture lacks capacity.



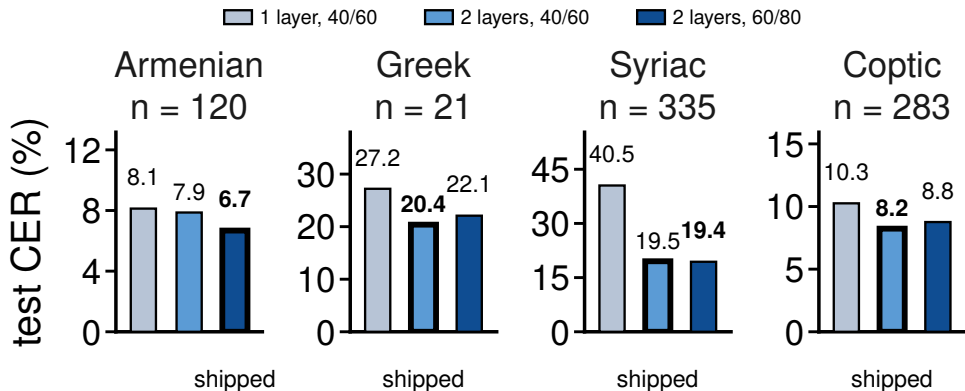


A second layer helps; a third adds no reliable gain.



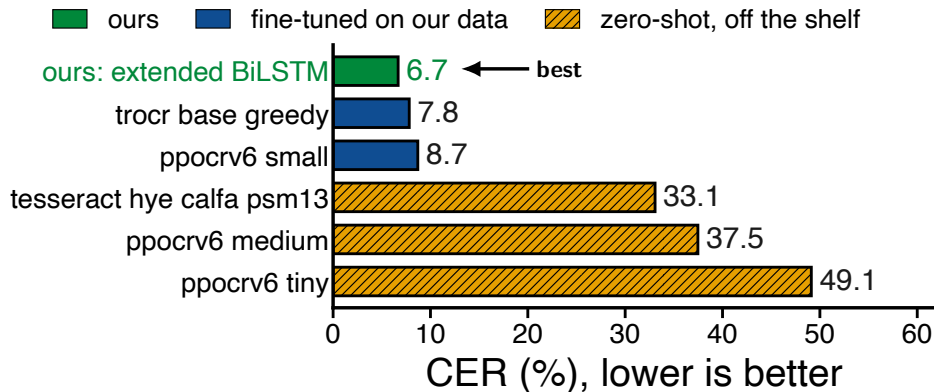
No width wins everywhere: Armenian wide, Greek and Coptic narrow, Syriac a tie.

The staircase: three steps per script



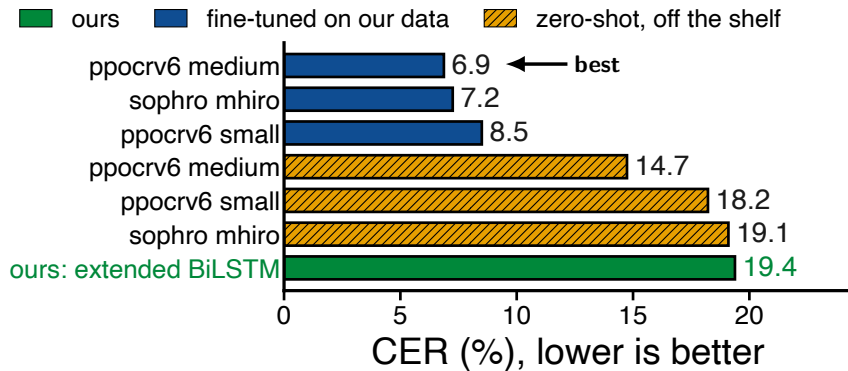
The second layer helps everywhere. The wider CNN only pays on Armenian, so only Armenian keeps it.

Armenian, n = 120 test lines



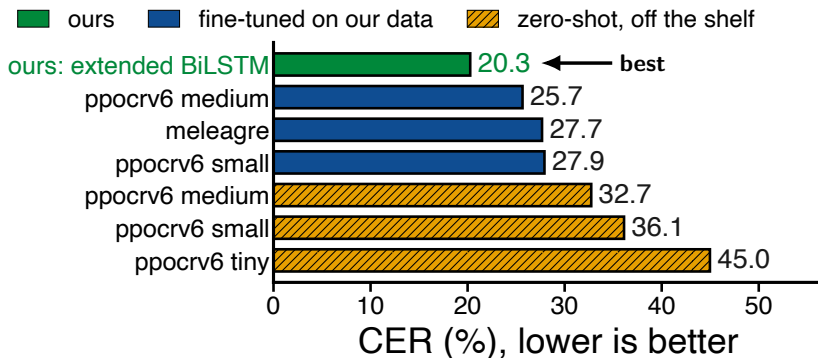
Ours **6.7%**. Best off the shelf 33.1%.

Syriac, n = 335 test lines



Fine-tuned PP-OCRv6 medium **6.9%**. Ours 19.4%. Even its zero-shot run beats us, at 14.7%.

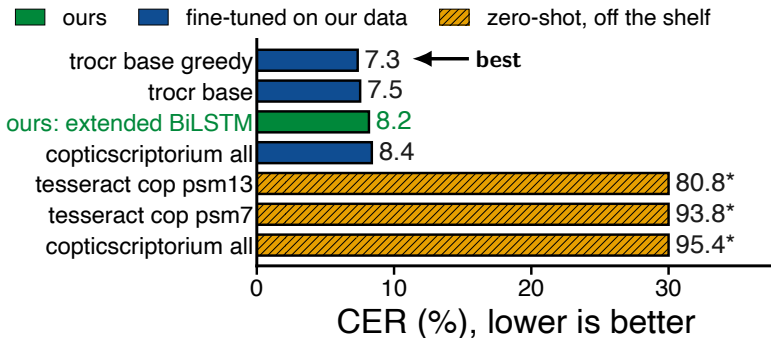
Greek, n = 21 test lines



ours here is the 60/80 checkpoint the benchmark report scored (NFD); the deck's Greek model is 40/60, at 20.4 CER

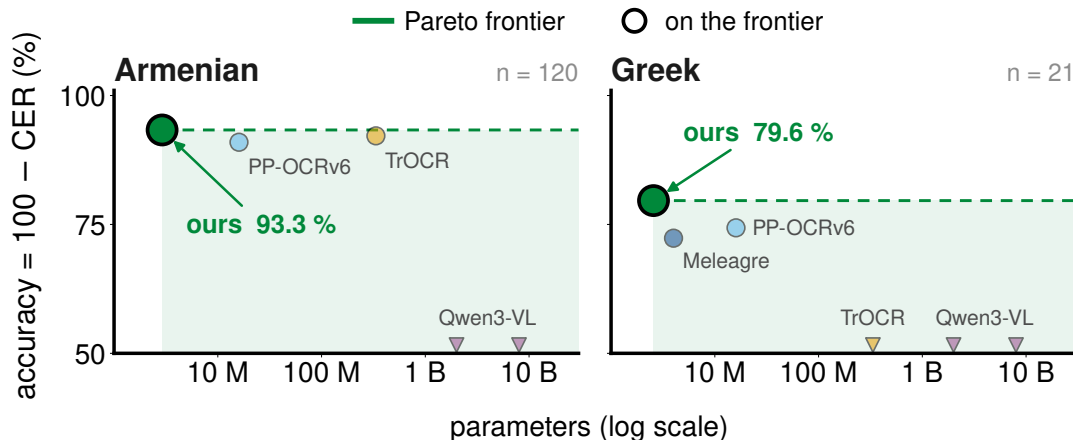
Ours **20.3%**. Best off the shelf 32.7%. Only 21 test lines.

Coptic, n = 283 test lines



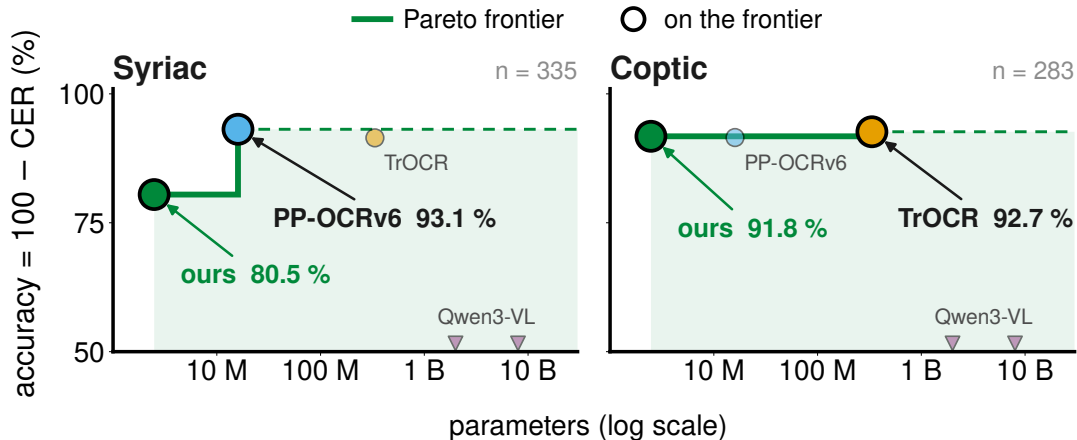
* bar cut at 30, true value shown

The one system that beats us is TrOCR base, fine-tuned on our own data: **7.3%** against our 8.2%.



Ours is alone on the frontier.

The Pareto frontier: Syriac and Coptic



Ours starts the frontier. Larger models take over.

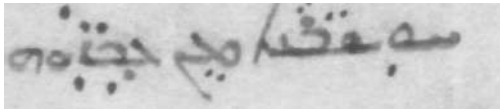
06

section 6 of 8

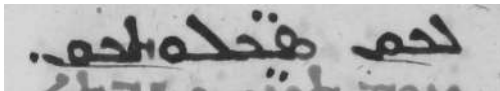
Error Analysis



1. Two densities



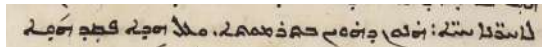
West, 2024 book: about 10 characters per 100 px



East, 2025 book: about 6 characters per 100 px

Calamari learns characters per 100 px. One ratio cannot fit both books.

2. Many small dots



Dots for letters, plurals and vowels



Our model drops the plural dots and the dot below

Many dots with different meanings, tiny at 48 px line height: hard for the CNN filters.

OeNB Cod. Syr. 1 (2024); HMML SMMJ 00036 (2025); dots: Vat. Borg. sir. 82

3. One book carries the errors

1,795 Syriac validation lines	West, 2024 book	East, 2025 book
Share of the lines	16%	84%
Share of the characters	34%	66%
Share of the character errors	60%	40%
CER	50.7%	17.5%

16% of the lines carry 60% of the errors.

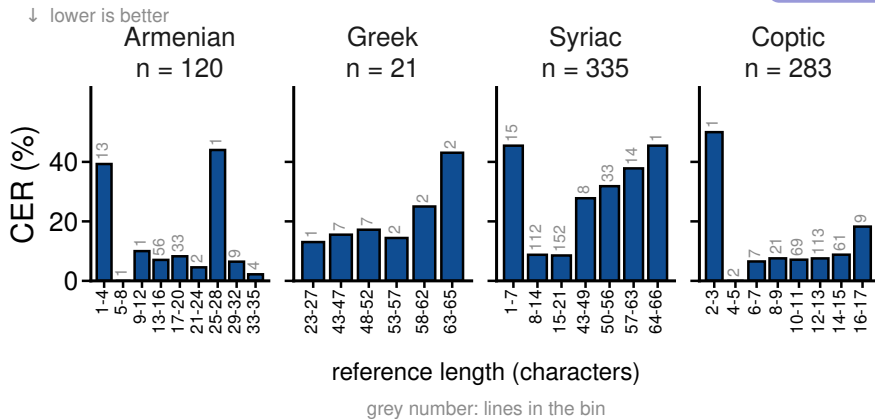
Open: rebalance per book, or ship PP-OCRv6 for Syriac.

On the same Syriac lines other models do better. Nothing is fixed yet.

Syriac pretraining validation lines, before the label cleanup, so these are not the test numbers on the earlier charts.

Short lines carry the worst error

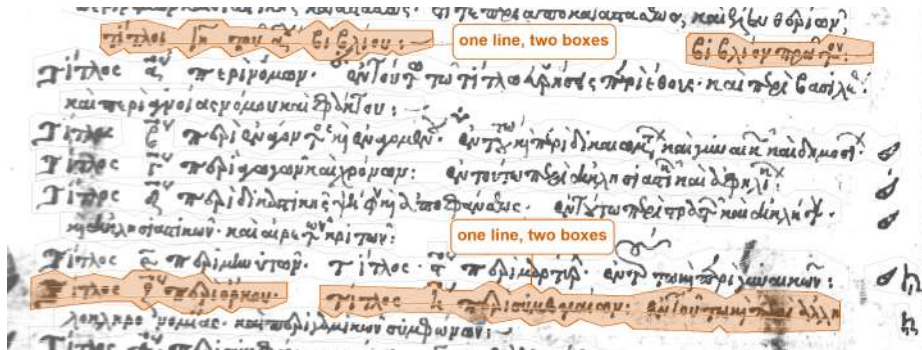
▶ what the short lines are



Gutter strips and folio numbers. Several bins hold only one or two lines.

Open: fix boxes and pairing

A bad crop cannot be read



Half the line never reaches the transcription model.

No model can recover a character that was cut off the crop.

Open: switch the segmenter

07

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Platform



The editor and the public link



The editor: draw, reshape and fix the boxes

Segment, transcribe, correct, share. At app.nomikos.app



The public view: read-only, no account, export

BnF Grec 1360



One corrected line, with its text

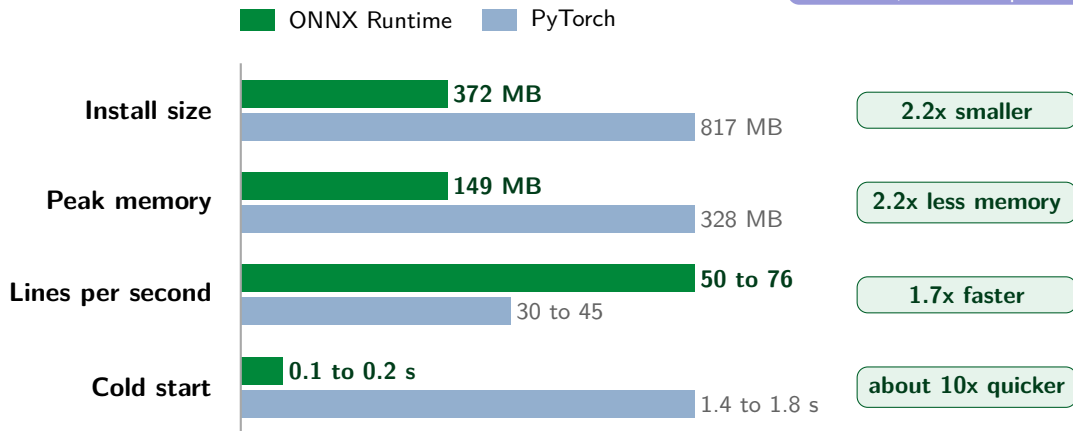


huggingface.co/nomikos-project

Checked lines are the next training set. The loop is built.

Why we export to ONNX

► the table, and the export check



Same weights, same output. One **9.6 MB** file per script reads a 30-line page in **0.5 s** on a laptop CPU.

7,174

lines segmented

143

pages in 17 documents

5,471

lines with expert text

6

active researchers

100,000

characters, about

205

jobs run



“NOMIKOS, in this respect at least, is a liberating development.”

“Whereas on Transkribus we must remember to draw the baseline in the segmentation process from left to right, the exact opposite of our text’s natural orientation, NOMIKOS offers us the freedom to work naturally on our manuscripts, without any trouble over left or right.”

Dr. Nima Jamali

Postdoctoral researcher, NOMOS project



Better than Transkribus, for segmentation and for transcription.

Dr. Federico Alpi's verdict after working with Nomikos on his manuscripts.

Postdoctoral researcher, NOMOS project

08

section 8 of 8

Conclusion

- What we found
- Limitations
- What comes next



1

Curated data

5,400+ lines with expert
text, in four scripts



2

Resource efficient models

2.5 to 2.9 M parameters,
running on a CPU



3

The platform

where researchers use
these models and
correct the output

Small per-script models match a **135x** larger transformer on three scripts. Syriac is behind.



One model for all three scripts

Syriac CER **19.4** → **32.9 %**

Greek 22.1 → 24.1 %,

Armenian 6.7 → 7.9 %

Confounded: its learning rate
did not match the baseline.



Reworking TrOCR for these scripts

Swapping its decoder or its
vocabulary made it worse.

No clean number, so we claim none.

First question for any pretrained decoder:
are the script's characters in its vocabulary?

Negative results count: nobody has to spend that compute again.

Where the numbers are thin

- Greek rests on 21 test lines: more Greek data needed
- One seed, no intervals
- Syriac needs more work: 19.4 % against **6.9 %**

What is still open

- Time saved per folio not measured
- Segmenter benchmark is Coptic only
- Some training manuscripts cannot be released publicly

Next models

- Train on the collected corrections
- Improve Syriac, or ship PP-OCRv6 for it
- TrOCR reads Syriac well, **7.9%** CER:
distil it into a small model that runs on a CPU

Then data

- Switch the segmenter
- Finish the label normalisation
- More Greek data, plus the pending experiments

The loop is built and the first corrections are collected; no model has been retrained on corrections yet.

Thank you

Questions or feedback?

Astha Gupta

`astha.gupta@campus.lmu.de`

Krrish Agarwalla

`krrish.agarwalla@campus.lmu.de`

Platform: `app.nomikos.app`

Models: `huggingface.co/nomikos-project`

- [1] Qoruyo: Models for automatic transcription of manuscripts. Beth Mardutho. URL <https://bethmardutho.org/qoruyo/>. Free, open-source kraken models for printed (Qoruyo) and handwritten (Sophro Mhiro) Syriac, Estrangela/Serto/East Syriac.
- [2] hye-tesseract: Open ocr model for armenian. GitHub, Calfa. URL <https://github.com/calfa-co/hye-tesseract>. Printed/noisy Armenian text, Classical/Western/Eastern Armenian, no accuracy figure published on the repo.
- [3] A text recognition dataset from sahidic coptic ancient manuscripts (scam). arXiv:2606.15987. URL <https://arxiv.org/html/2606.15987>. CC BY 4.0; 3,240 lines, best VAN baseline 5.25% CER.
- [4] kraken-ppocrv6-medium. Hugging Face. URL <https://huggingface.co/small-models-for-glam/kraken-ppocrv6-medium>. Apache-2.0; 44 languages / 10 scripts incl. Armenian, Greek, Syriac (not Coptic); 3.91% aggregate CER.
- [5] Greek_medieval-and-modern-minuscule. Transkribus public model, ID 45032. URL https://www.transkribus.org/models/greek_medieval-and-modern-minuscule. 2.4% CER; 1,051 pages / 32,481 lines from Bodleian, British Library, and Cambridge collections.
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- [12] Minghao Li, Tengchao Lv, Jingye Chen, Lei Cui, Yijuan Lu, Dinei Florencio, Cha Zhang, Zhoujun Li, and Furu Wei. TrOCR: Transformer-based optical character recognition with pre-trained models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 13094–13102, 2023. URL <https://ojs.aaai.org/index.php/AAAI/article/view/26538>.
- [13] Sourab Mangrulkar, Sylvain Gugger, Lysandre Debut, Younes Belkada, Sayak Paul, Benjamin Bossan, and Marian Tietz. PEFT: State-of-the-art parameter-efficient fine-tuning methods, 2022. URL <https://github.com/huggingface/peft>.
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- [16] Christian Reul, Uwe Springmann, Christoph Wick, and Frank Puppe. Improving OCR accuracy on early printed books by utilizing cross fold training and voting. In *13th IAPR International Workshop on Document Analysis Systems (DAS)*, pages 423–428. IEEE, 2018. doi: 10.1109/DAS.2018.30. URL <https://doi.org/10.1109/DAS.2018.30>.
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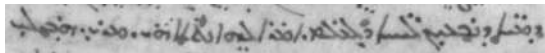
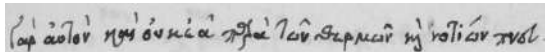
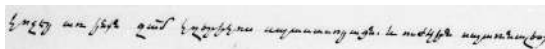
- Segmentation: kraken¹¹.
- Transcription model architecture: Calamari¹⁹.
- Training and export: PyTorch¹⁵, ONNX Runtime¹⁴.
- Transformer runs: Transformers²⁰, PEFT¹³.
- Tracking and configuration: Weights and Biases⁶, Hydra²¹.

A

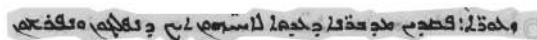
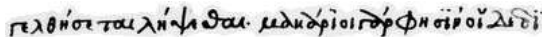
Appendix

- Data and labels
- Segmentation
- Model and experimental setup
- Results: our levers
- Results: the benchmark
- Error analysis
- Platform and reference

Pretraining: other manuscripts, printed editions, public datasets



Finetuning: the manuscript the researchers care about



Pretraining lines teach the script, finetuning lines teach the hand.

Dulaurier; Eparchos; Syriac printed book; MS P 331; BnF Grec 1360; Syriac Chapter 4

Pretraining



low contrast



faded page



ink blob

Finetuning



ink fades out



folio number



dots, not text

Both sets contain bad lines.

Byzantine Greek

- EPARCHOS: about 120 pages of Renaissance Greek hand, c. 1500 to 1530
- Stavronikita 53, 79 and 114: Gospel codices, Mount Athos
- Codex Palatinus graecus 23: 10th-century Palatine Anthology, Heidelberg
- HPGTR: Bodleian Barocci palaeographic Greek, 70 pages with ALTO
- LJS 380: 15th-century Penn manuscript, 104 line crops
- Dresden SLUB A.151: Byzantine Greek, XML and text
- Bessarion: 37 painted donor inscriptions in Epirus

- New Testament 1550: **printed**, not handwritten
- Esteban: binarised line crops derived from several of the above

Armenian

- Calfa and BnF Dulaurier: 42 microfilm images, 1,467 lines
- MS P 172: canon law, 17 colour pages, Transkribus export
- UCLA Ms 72: codex excerpt ff. 64v to 73v, 17 colour pages
- MS P 331: third Armenian manuscript, ff. 24r to 34r

Syriac

- 2024: OeNB Cod. Syr. 1, 140 folios, Serto, Vienna 1545
- 2025: HMML SMMJ 00036, 133 bifolios, mainly Estrangela
- Chapter 4: Vatican, the finetuning manuscript

Coptic

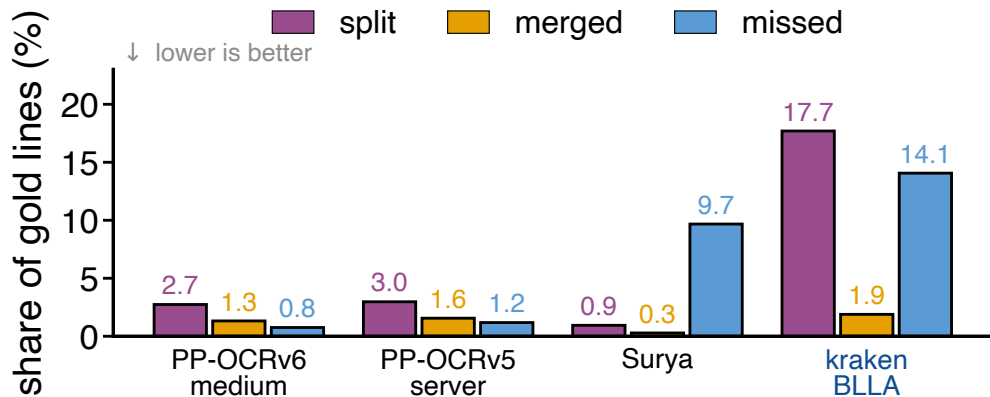
- Vatican Copt. Borg. 109 and Napoli IB.10.10-27
- SCAM, Sahidic Coptic: annotations only, images withheld

Licence to be confirmed for every source above: the repository records no licence for any of them.

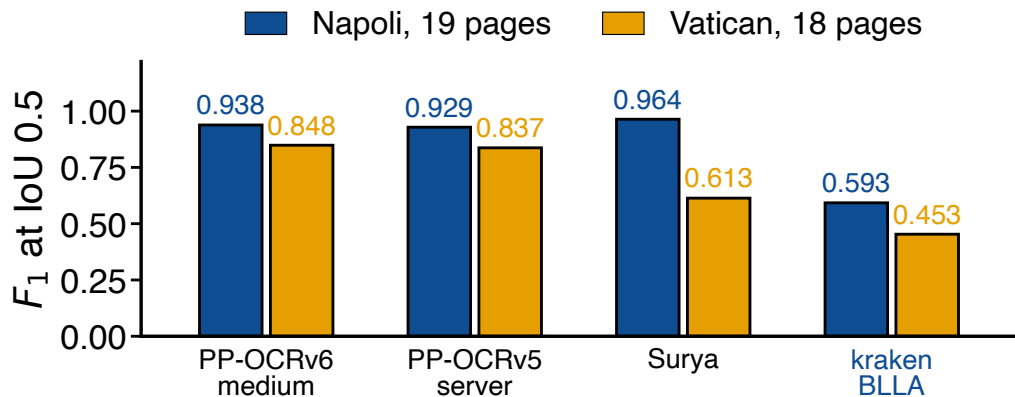
The project's own note: the Greek sources "Esteban" and "Dresden A151" have no documented provenance or licence.

- Licences differ per holder, and the repository records none
- Greek pretraining provenance unknown
- Report ships a code supplement

The previous frame lists every source we hold. Tracing and citing the Greek pretraining sources is still open.



Splitting, not missing: one written line comes back as two boxes.

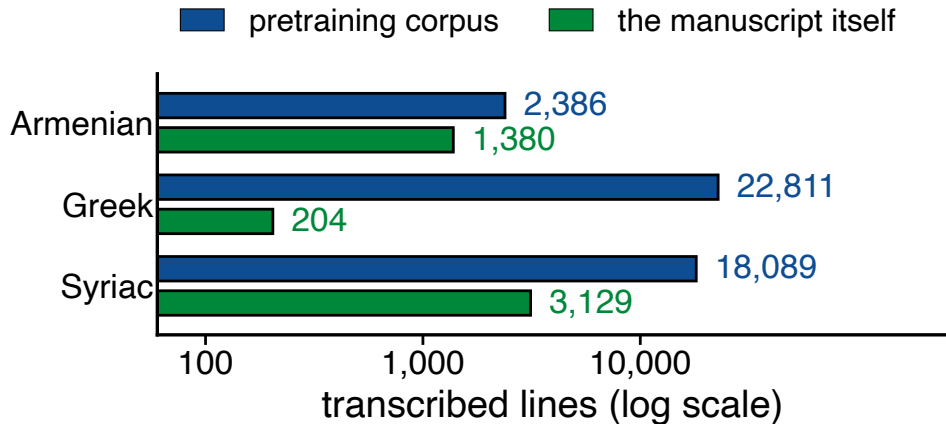


19 Napoli openings and 18 Vatican folios: two columns, very little data behind each.

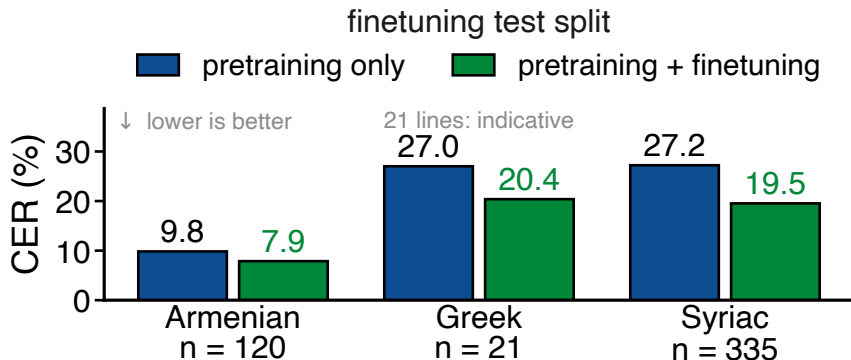
Manual-only gold	$F_1@0.5$	$F_1@0.75$	split %	miss %	false %	s/page
PP-OCRv6 medium	0.913	0.434	2.7	0.8	10.3	0.06
PP-OCRv5 server	0.903	0.309	3.0	1.2	10.0	0.07
Surya	0.872	0.549	0.9	9.7	6.6	0.41
kraken BLLA	0.548	0.260	17.7	14.1	18.7	15.31
CRAFT [†]	0.393	0.146	0.7	5.0	44.0	0.18
docTR [†]	0.345	0.108	1.5	17.2	36.9	0.03

[†] word detectors, not line segmenters: their words merged into lines, wide gaps cut. split % and miss % are shares of gold lines, false % of predictions.

- Zero-shot, public weights. Nothing was fine-tuned on Coptic.
- 37 pages, two manuscripts, one script.
- Gold: the 2,119 lines a person drew. The kraken-sourced gold lines are excluded, because scoring on them rewards kraken.
- PaddleOCR returns quadrilaterals; kraken, Surya and the gold are polygons, so IoU is not comparable across that line.
- Intervals are a paired bootstrap over the 37 pages.



The gold set is the small one: 204 Greek lines against 22,811.

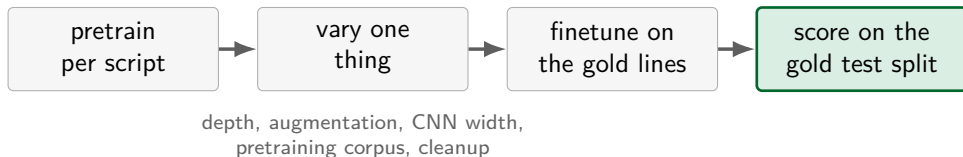


Left bar: pretraining only. Right bar: pretraining + finetuning.

Line height	48 px
Optimiser	AdamW, wd 0.01
Warmup ratio	0.09
Pretrain	lr 10^{-3} , 150 ep
Finetune	lr 10^{-4} , 100 ep
Batch size	32
EMA decay	0.99
Augmented copies	5 per line
Decoding	greedy CTC
Seed	1111

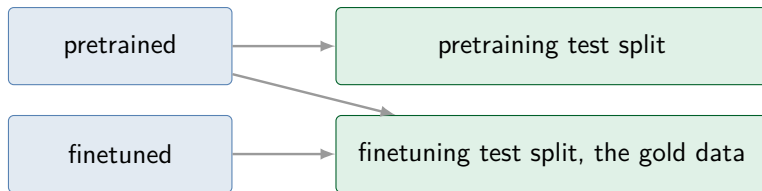
Deployed models: 40 and 60 convolution filters, two BiLSTM layers of 200 units per direction, dropout 0.3, about 2.5 million parameters per script.

The Armenian finetuning learning rate was swept to 9.94×10^{-5} and is reported as 10^{-4} .



One run per setting, seed 1111. Checkpoint chosen on validation CER only; the test split is scored once per checkpoint.

Which lines each model is scored on



	Pretraining lines			Finetuning lines		
	train	val	test	train	val	test
Byzantine Greek	18,239	2,298	2,274	164	19	21
Armenian	1,807	297	282	1,141	119	120
Syriac	14,499	1,791	1,799	2,463	331	335
Coptic		none		2,258	282	283

- Fixed split files `gt_train/val/test.txt` per language and stage. Every study scores the same file.
- Pretraining: 150 epochs, lr 10^{-3} . Finetuning: 100 epochs, lr 10^{-4} , initialised from the pretrained `best.pt`.
- Checkpoint selection on validation CER, top one only. Test scored once per checkpoint.
- Greedy CTC decoding. No beam search, no language model, no dictionary, anywhere in the project.
- Metrics are corpus-level: CER, WER, exact match.
- Test split sizes. Finetuning: Armenian 120, Greek 21, Syriac 335. Pretraining: Armenian 282, Greek 2274, Syriac 1799.

- Per-script byte-level BPE tokenizer
- Decoder re-initialised
- LoRA adapters on the encoder

TrOCR¹², LoRA¹⁰.

- No lexicon exists for these scripts
- Beam search gains little on CTC
- Voting multiplies size and latency

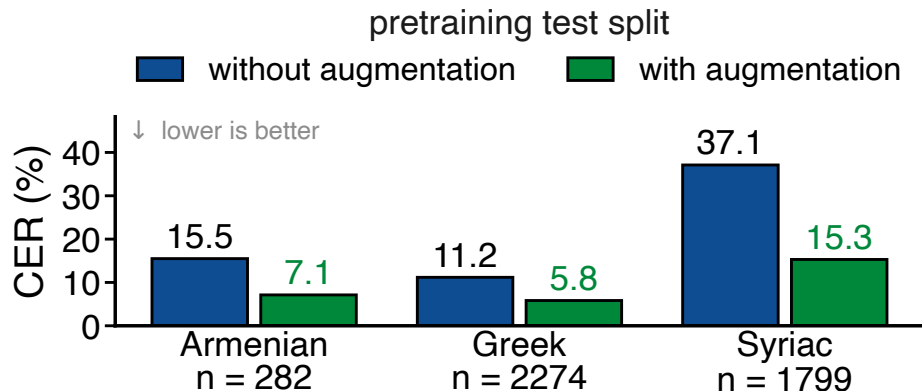
Cross-fold voting reduces error on early printed books¹⁶. We never measured beam search against greedy on our own lines.

CER (%), finetuning test split	Armenian $n = 120$	Greek $n = 21$	Syriac $n = 335$
Pretrained	9.83	27.01	27.23
Finetuned	7.87	20.38	19.53
1 BiLSTM layer	8.13	27.20	40.50
2 BiLSTM layers	7.87	20.38	19.53
3 BiLSTM layers	7.30	20.09	20.13

Depth rows are finetuned checkpoints, 40/60 filters.

CER (%), finetuning test split	Armenian $n = 120$	Greek $n = 21$	Syriac $n = 335$
Without augmentation	13.56	32.04	41.38
With augmentation	6.68	22.09	19.38
40/60 filters	7.87	20.38	19.53
60/80 filters	6.68	22.09	19.38

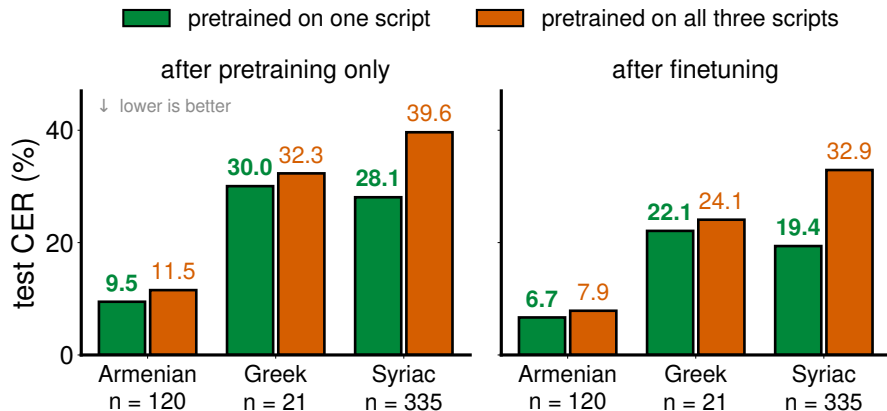
Augmentation rows use 60/80 filters, depth rows in the previous table use 40/60. Shared checkpoints agree to full precision.



- Augmentation probability is 1.0 with five extra copies, so an augmented epoch sees six variants of every line.
- The two arms are matched on epochs, so the augmented arm also takes about six times as many gradient steps.
- No steps-matched control run exists, so part of the gain is extra training rather than extra variety.
- The effect is much larger than any capacity effect we measured, so the main chart's ordering survives the caveat.

CER (%)	Armenian	Greek	Syriac
<i>finetuning test split</i>			
Monolingual pretraining	6.68	22.09	19.38
Combined pretraining	7.87	24.08	32.92
<i>best-epoch validation, 3 BiLSTM layers</i>			
Before cleanup, finetuned	8.97	15.93	28.20
After cleanup, finetuned	9.16	15.62	18.12
Before cleanup, pretrained	13.99	6.50	28.74
After cleanup, pretrained	11.85	5.76	18.98

The cleanup block is validation, not test, and the two sides were scored against different label files.



One model per script wins everywhere. Syriac: **19.4%** against 32.9%.

Fine-tuned test CER	One script (monolingual)	All three scripts (combined)	Change
Armenian	6.68 %	7.87 %	↑ 1.19 pp worse
Greek	22.09 %	24.08 %	↑ 1.99 pp worse
Syriac	19.38 %	32.92 %	↑ 13.53 pp worse

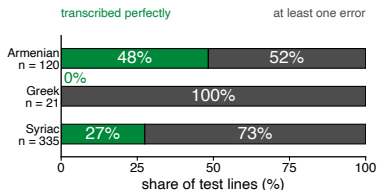
Pretraining one model on all three scripts lost on every script, and most on Syriac.

Same data, augmentation, seed and epoch budget. Not matched: the pretraining learning rate, see the next frame.

- Combined pretraining used $\text{lr } 10^{-4}$, monolingual used 10^{-3} . That is the one unmatched hyper-parameter.
- The combined runs also see roughly 1.9 to 19 times more gradient steps at matched epochs.
- Finetuning lr is matched at 10^{-4} , but a finetuned combined row inherits the confound from its pretrained parent.
- The study's own notes ask for a rerun at matched learning rate. Until then: "did not help in our setup", not "does not work".

- Every comparison in the talk is pretrained against pretrained plus finetuned
- The control we never ran: train on the few hundred gold lines alone, from scratch
- That run is requested. The two-bar chart, finetuning only against pretraining plus finetuning, belongs here

Finetuning test split	Armenian <i>n</i> = 120	Greek <i>n</i> = 21	Syriac <i>n</i> = 335
CER (%)	7.87	20.38	19.53
WER (%)	37.35	64.46	53.29
Exact match (%)	36.7	0.0	27.5

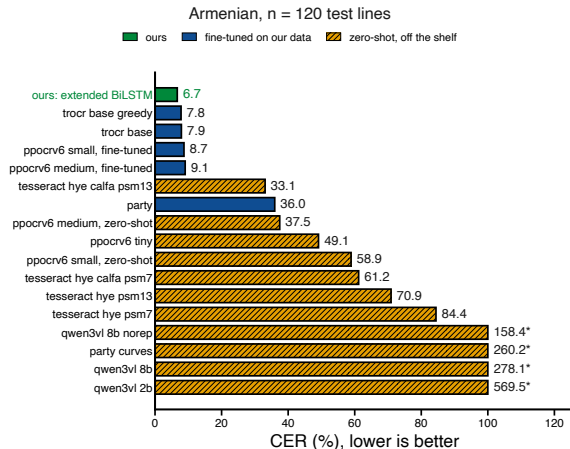


Greek scores zero exact match on all 21 lines. The chart is the width we ship per script, so Armenian reads 48%; the table above is the 40/60 checkpoint.

Script	TrOCR base, fine-tuned	Ours	Test lines
Armenian	7.8%	6.7%	120
Greek	70.6%	20.3%	21
Syriac	8.5%	19.4%	335

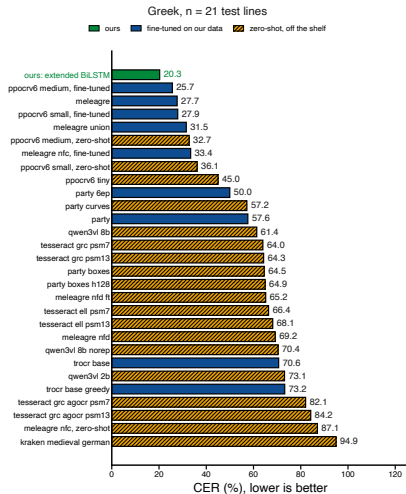
Corpus-level CER on the finetuning test split, the same lines for both. TrOCR base is 334 M parameters against our 2.9 M.

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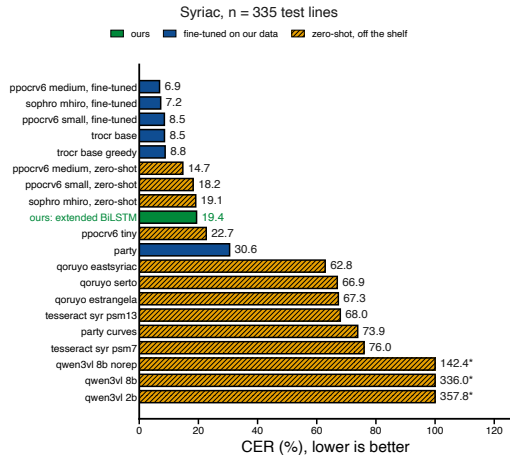


* bar cut at 100, true value shown

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* bar cut at 100, true value shown

- One run per setting, seed 1111. No repeated seeds, no confidence intervals anywhere.
- A few tenths of a point is noise. Two trainings of the same setup landed 0.7 to 2.2 points apart.
- Greek finetuning is 21 test lines and 19 validation lines. One test line is about 4.8 points of exact match.
- CER is not comparable across languages: different corpus sizes, scripts and label quality.
- Split files are fixed on disk, but how they were built is not documented and no manifest hash is committed.
- Bootstrap intervals over test lines are cheap to add and need no retraining, but per-line predictions were not saved.

- A transcription model per script, trained from a few hundred gold lines
- A curation and pairing pipeline that replaces most of the manual work
- Serving that reproduces the training loader exactly
- A runtime that fits a laptop CPU

Four scripts, a few thousand lines each, and one laptop. No published pipeline we found does all four for these scripts.

The short tail is folio numbers and gutter strips

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54 Armenian crops of one to three characters; 125 of 159 flagged Syriac crops were not the margin numeral.

- A codec is built from its own training split, so a pretrained checkpoint can meet a character it cannot encode.
- Three Syriac vowel marks (U+0735, U+0738, U+0739) and one Greek iota with macron (U+1FD1) are affected.
- The evaluator drops such a character from the CTC target only. The reference text is untouched, so CER, WER and exact match stay exact.
- One Syriac crop is shorter than its target after downsampling, so CTC returns an infinite loss and the batch mean goes infinite. No report table has a test-loss column.

Transcribe, batch 1	ONNX Runtime	PyTorch
Install size	372 MB	817 MB
Peak memory	149 MB	328 MB
Lines per second	50 to 76	30 to 45
Cold start	0.1 to 0.2 s	1.4 to 1.8 s
Model artifact	about 10 MB per script	

Line. One line of writing: a baseline and a polygon.

Line crop. The image cut around that polygon.

Ground truth. The expert transcription after curation.

Pairing. Matching a segmented line to its text.

Pretraining set. Lines that teach the script.

Finetuning set. Curated lines from the target document.

HTR. Reading one handwritten line. OCR is the umbrella word.

CER. Wrong characters per character of ground truth.

WER. The same, over whitespace-separated tokens.

Exact match. Share of lines that match character for character.